

Experiment-3

I-V Characteristics of Schottky Diode

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Course :M.Tech VLSI &ES

Aim: Design Schottky diode and study the I–V characteristics using Synopsys TCAD Sentaurus.

Objectives:

1. Compare I-V characteristics of Schottky and PN diode.
2. Study the effect of doping concentrations of Schottky diode.

Software Tool:

Synopsys TCAD Sentaurus.

Theory:

A Schottky junction is a metal-semiconductor interface that exhibits rectifying behavior, allowing current to flow more easily in one direction. It typically involves a metal like titanium and a semiconductor such as silicon. Schottky diodes are known for their fast-switching speeds due to minimal charge storage.

Procedure:

1. Create a new folder. Open in Terminal in new folder then enter “sde “command in the terminal.
2. Go to View ☐ GUI Mode ☐ 2D Mode.
3. Go to Draw then deselect the “Auto Region Naming “and select the “ExactCoordinates”.
4. Now go to materials and select “silicon” (silicon is the default material) then click on “Create rectangular Sheet Region” on toolbar.
5. Place the rectangle on Canvas with required coordinates and press ok.
6. Now enter Region Name as “nregion”.
7. Go to View -> Grid and change Grid Width and Grid Height values to 0.003.

8. Now create the p region inside the n region for that select Snap to Grid from Tool Bar.
9. Select the Rectangle, draw the p region with required coordinate values.
10. For doping the nregion, Go to Device -> Constant Profile Placement. Select the region as nregion and change the parameters as required.
11. For doping the pregion, Go to Device Constant Profile Placement. Select the region as pregion and change the parameters
12. For creating Contact, Go to Contacts ☐ Contact Sets. Enter the contact name for pregon and nregion with different edge colour.
13. To Set Contact for pregon and nregion first change the Select Body to Select Edge on Tool Bar.
14. Now click on Arrow () placed at left panel and go to Contacts -> Set Contact then place the electrical contact on the edge of pregon and nregion.
15. Go to Mesh -> Define Rel/Eval Window and select rectangle -> draw rectangle on canvas covering the entire design.
16. Got Mesh -> Refinement Placement enter the minimum & maximum element size. Click on create Refinement.
17. Go to Mesh -> click on Build Mesh. Then the device structure will be open in the svisual.

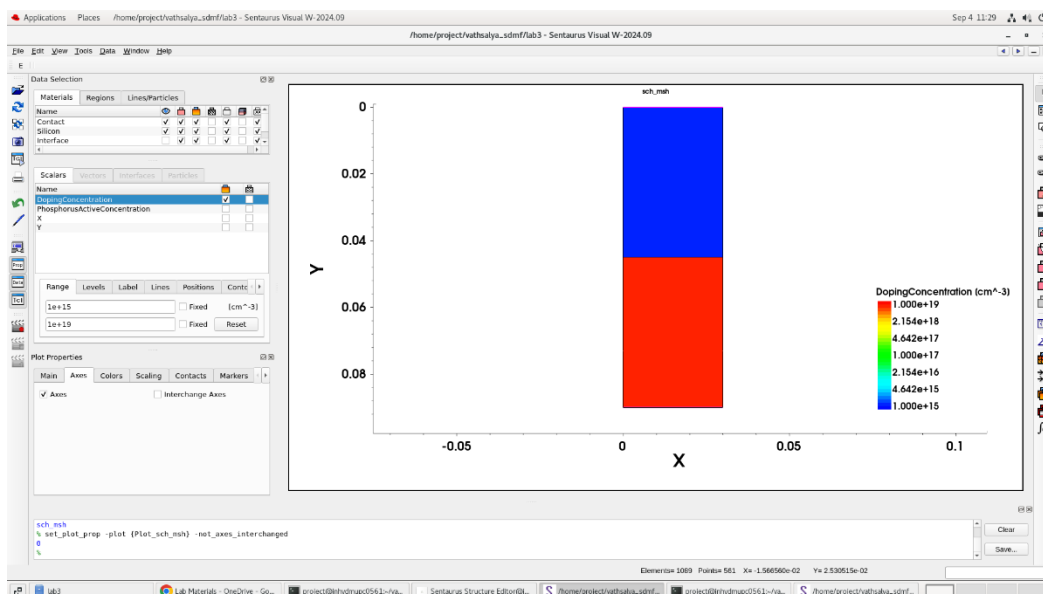


Fig 1: Mesh for doping concentration $1e15$ to $1e20$

18. To study the device characteristics, edit the diode "cmd file" (.cmd file). msh.tdr file name should be included in cmd file Grid section.

```
File{
  Grid      = "schottky_msh.tdr"
  Plot      = "silicon"
  Current   = "silicon"
  Output    = "silicon"

}

Electrode{
  { Name="pcontact" Voltage=0.0 }

  { Name="ncontact" Voltage=0.0 }

}
```

19. Enter the "sdevice cmdfilename" in the working directory terminal.
20. To visualize the device characteristics, enter **svisual** command in terminal.
21. Go to file in svisual and open plot file.
22. For plotting I-V characteristics select pcontact Outer Voltage on x-axis and pcontact Total Current on Y-axis.

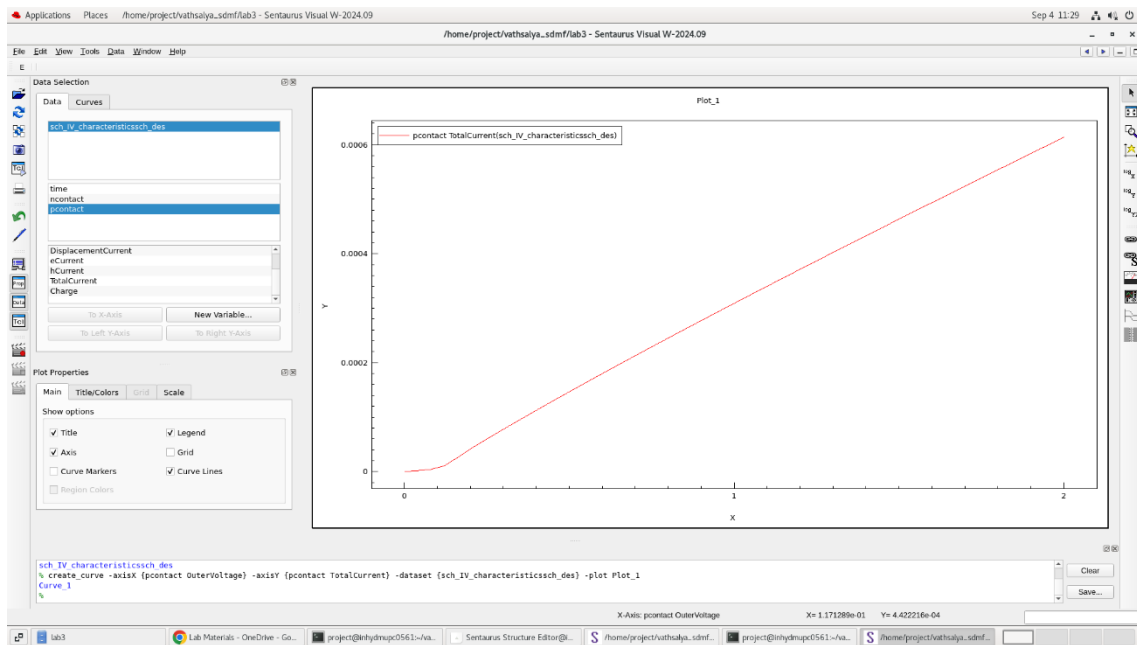


Fig 2: I-V characteristics plot

23. Now change the doping concentration of pregion to $1e15$ and nregion to $1e21$ then plot the I-V characteristics for both conditions.

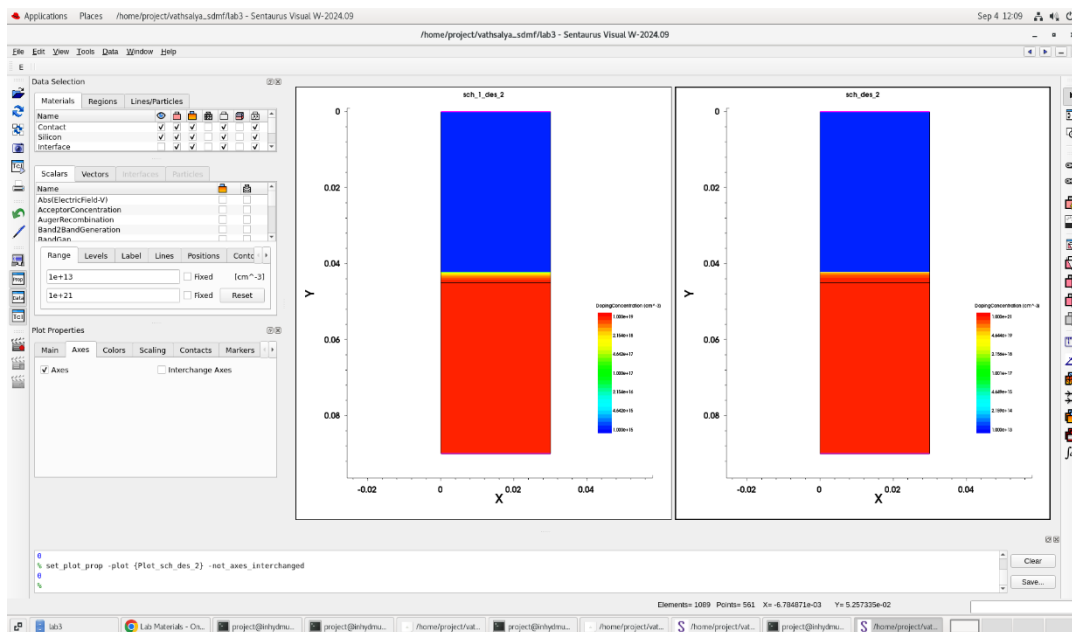


Fig 3: Mesh for doping concentration $1e15$ to $1e21$

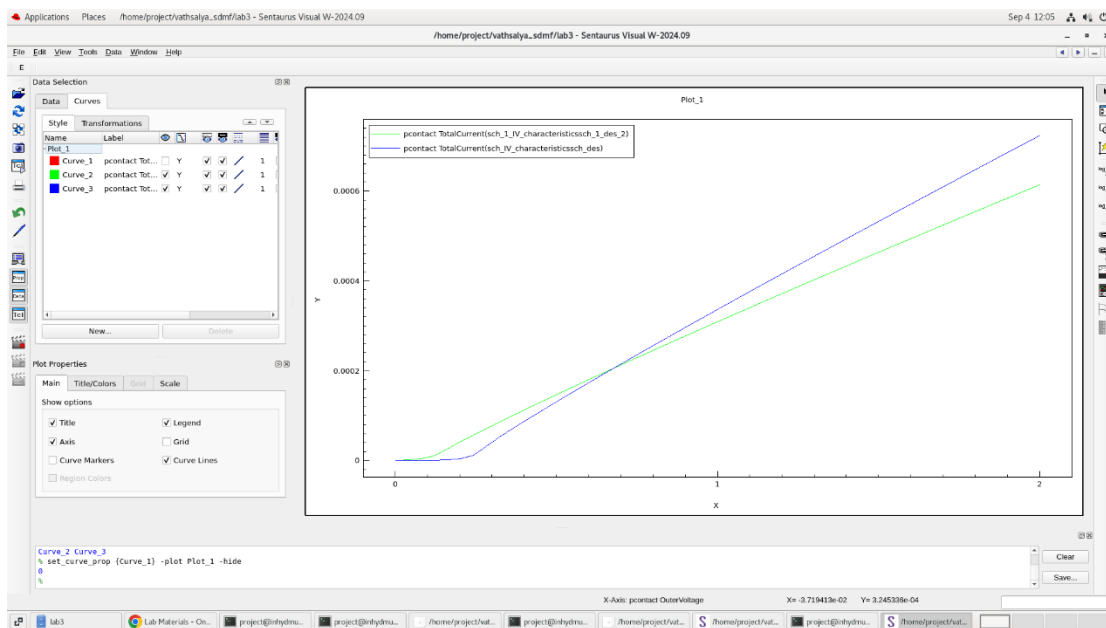


Fig 4: I-V characteristics plot for different doping concentrations

24. Now change the pregion from silicon to metal (Gold) without doping concentration and nregion doping concentration to $1e+16$

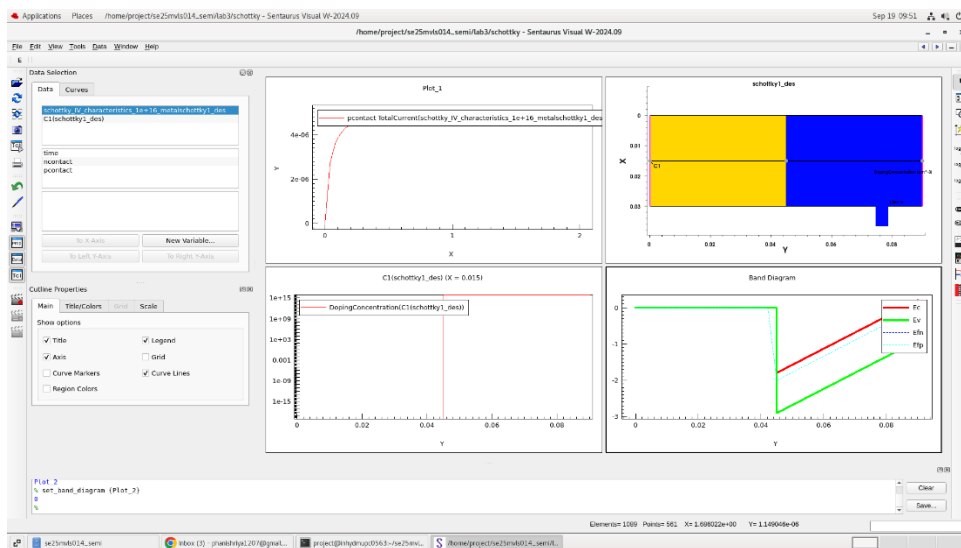


Fig 5: Mesh

25. After that plot the I-V characteristics select pcontact Outer Voltage on x-axis and pcontact Total Current on Y-axis.

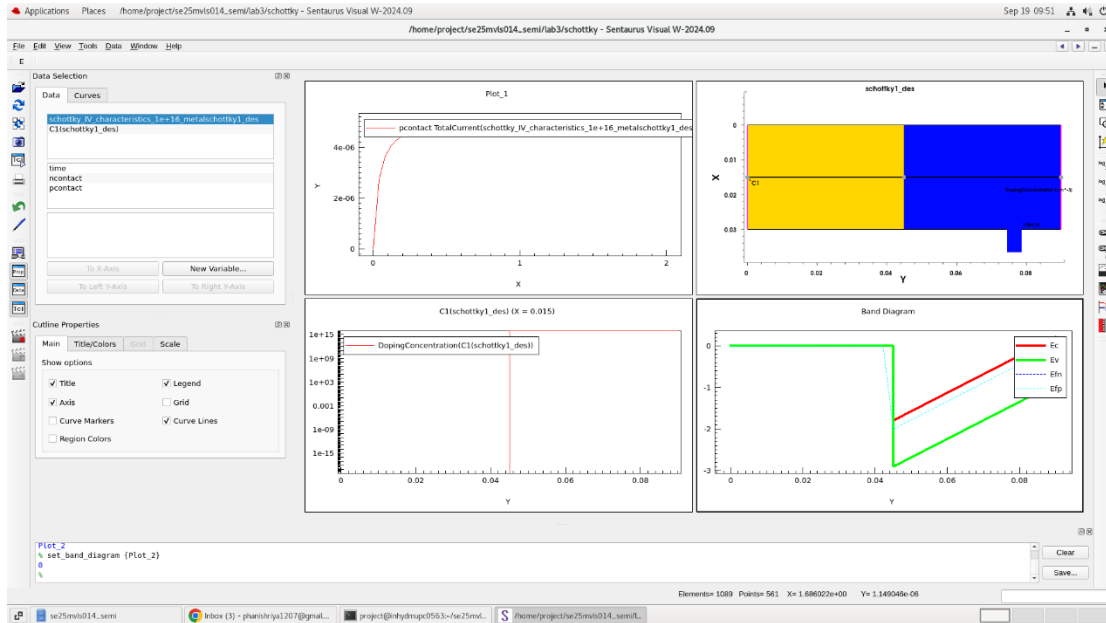
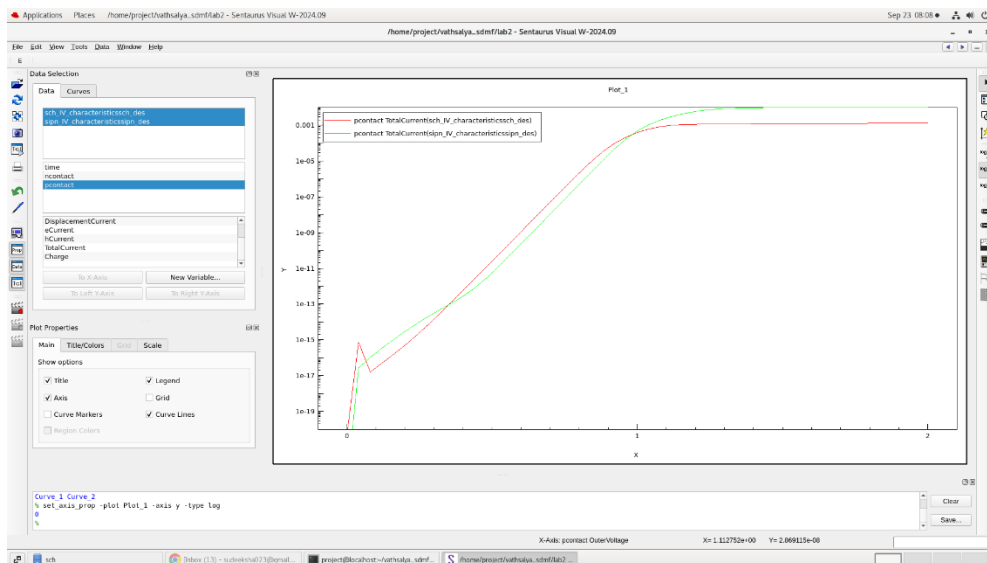


Fig 6: I-V characteristics plot

Fig 7 Comparison of PN diode and Schottky



Conclusion: The I-V characteristics of Schottky diode are studied using Synopsys TCAD Sentaurus software.